

PHY 2048 – Section 0007

General Physics Using Calculus I

Introduction and Learning Guidelines

Tuesday & Thursday 12:00 PM - 1:15 PM



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Department of Physics
University of Central Florida

Aug 25, 2026

Why cats always land on their feet?



Why do we study physics?



Daytona International Speedway



Why do we study physics?



Rockets 101 | National Geographic



Tacoma Bridge Collapse



Syllabus Review

- For syllabus, please visit: WebCourses@UCF (Simple syllabus)
(<http://webcourses.ucf.edu>)
- For digital “**iclicker**” subscription, please visit:
<https://join.iclicker.com/LTAT>
- Please use your UCF email ID to create account in “**iclickers**”.
(**Be careful: the free trial expires after 15 days**)



About the instructor

- Dr. Manjita Shrestha, Lecturer
- **Office: Physical Science Building, Room # 321**
- **E-mail:** manjita.shrestha@ucf.edu
- Communications by email is preferred. Responses to messages sent by Webcourses Inbox may be delayed.
- Office hours or free tutoring sessions: **Mondays & Wednesdays** 10:00 AM – 10:59 AM. (Other times are available by appointment).

Learning Modules vs Textbook

1. **Difference between learning modules and textbook:**

- Learning Materials in the Modules $\approx$ Textbook. Textbook has more details, while learning modules have extracted the most important contents from the textbook and modified them into a self-learning-evaluation modular system.

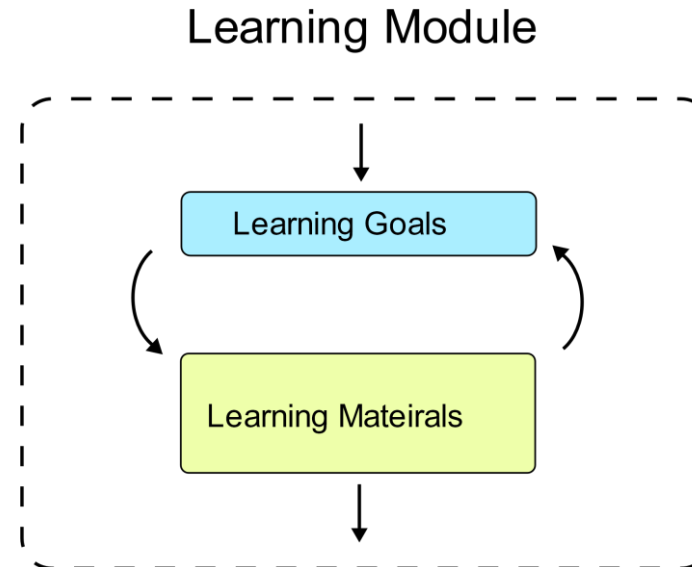
2. **Which one should I start first, textbook or modules?**

- Depends on your background knowledge and learning habit.
- If you already know some knowledge and are confident, you can work on the learning modules first and then return to learning materials or textbook as needed.
- If you don't have any knowledge yet, it is the best that you read through textbook first to learn concepts/physics laws.

3. **Lecture slides:** Summary of key concepts, knowledge, and example problems to make sure everything is covered.

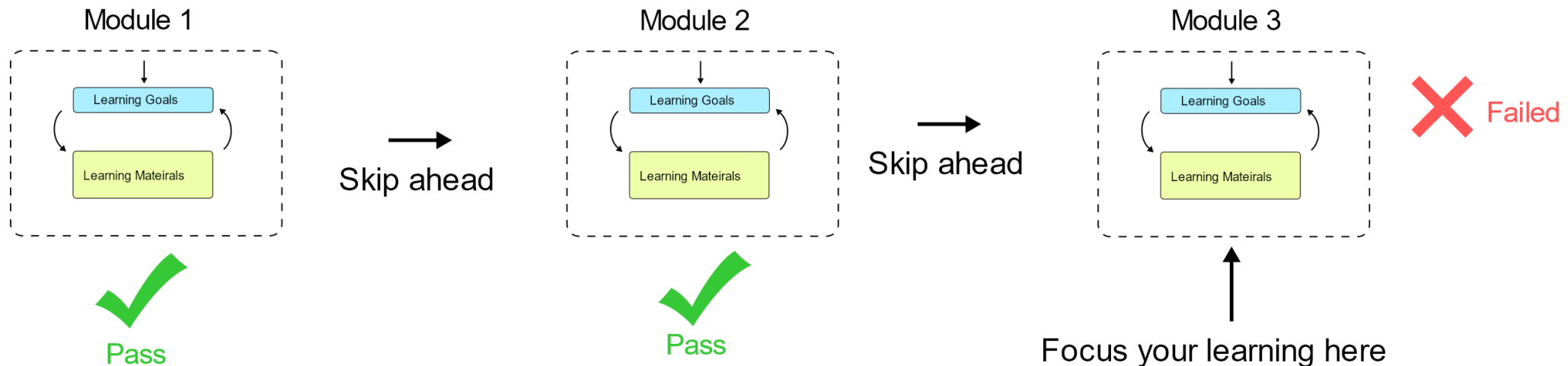
How to use Online Learning Modules

- Each learning module will consist of two parts: *Learning Goals* and *Learning Materials*.
- **Learning Goals** are a set of 1~2 problems, which serve as the assessment of the module. You will have multiple attempts at those Learning Goal problems. You can think of them as "level bosses" in a game.
- **Learning Materials** contain learning materials and practice problems designed to train you on mastering the concept and solving the "learning goals" problems, as well as similar problems on future exams.



Completing the Learning Modules

1. You are allowed and required to make one "free shot" at the Learning Goal problems. If you pass, you will immediately unlock the next module. If not, you'll have access to all the Learning Materials in the module to help you learn to solve this type of problem.
2. After the initial "free shot", you'll have two more attempts to get 100% on the current module and move on to the next one. If you fail to get 100%, you'll have two more attempts (attempts 4 and 5), but you will only get a maximum of 90% on this module even if you did all the problems correctly.
3. If you cannot pass the module in all 5 attempts, your highest score will be sent to the grade book, and you can move on to the next module. It is a good idea to ask for additional help in that case.



Learning Physics at Three Levels

1. Physics Concepts and Definitions

Displacement, velocity, speed, acceleration

2. Physics Laws and Formulas

1D motion kinematics equations (constant a)

3. Use Formulas to Solve Problems

Constant acceleration motions, free fall motion

Time Management

- Management of time will be essential if you are to keep up with the **fast pace of the course**. Since the fundamental concepts of physics being covered in this course iteratively build upon themselves, you mustn't **allow yourself to fall behind**.
- To succeed, it is estimated that you will need to allocate the following efforts for this course:
- Pre-class reading of the corresponding textbook – ... hours
- Lecturing, learning modules, and interactions – ... hours
- Recitation Activity – ...hour
- Completing homework assignments – ... hours
- Total estimated weekly effort – ...hours

Time Management

- If you do not understand something covered in the previous week, you will not have the foundation necessary to understand what is being taught during the current week.
- Do not let yourself fall into this trap, otherwise your situation will become increasingly hopeless the longer you put things off.
- Do not assume that simply reading the textbook or slides is enough. Until you can consistently work through problems, you do not understand the material well enough.

Tips for success

The following expectations may help you succeed in this course. Please carefully review these expectations and follow them.

- 1. Check your email every day or log into Webcourses several times each week** to check the course content, announcements, conversations, and discussions.
- 2. Keep up with the weekly readings and assignments.** Students who keep up with the weekly reading and assignments tend to do much better in the course than those who do not.
- 3. Please do not miss an assignment deadline.** Refer to the course schedule to ensure that you submit assignments on time.
- 4. Communicate with the instructor** and the TA on time, about your questions, concerns, and difficulties. We will do our best to help, support, and accommodate.

Exams at the Evaluation and Proficiency Center (EPC)

EPC Exam Instructions:

- Please follow the steps below to schedule your quizzes and exams through EPC:
- **Access EPC:** Use the [EPC - Home](#) link, or search for **EPC at the University of Central Florida** on Google.
- **Sign In:** Sign in using your **UCF account and password**.
- **Schedule an Appointment:** Schedule an appointment through EPC for the EPC Introduction Quiz and exam.
- **Accommodations:** If you would like to take an exam at the **SAS Center**, please schedule an appointment with the SAS Center in advance for your exams.

Please make sure that EPC Introduction Quiz and exams are scheduled ahead of time and that you follow the applicable UCF procedures.

EVALUATION AND PROFICIENCY CENTER

12760 Pegasus Drive
Orlando, Florida, 32816-1400
Room: ENGR2-201

☎ **(407)823-5619**

@ **epc@ucf.edu**

Hours

Tuesday	9:00 AM - 9:00 PM
Wednesday	9:00 AM - 9:00 PM
Thursday	9:00 AM - 9:00 PM
Friday	9:00 AM - 9:00 PM
Saturday	Closed
Sunday	Closed
Monday	9:00 AM - 9:00 PM

Thank you for your attention!

**I am looking forward to a successful
and productive semester together.**

PHY 2048

Units & Vectors

Main points

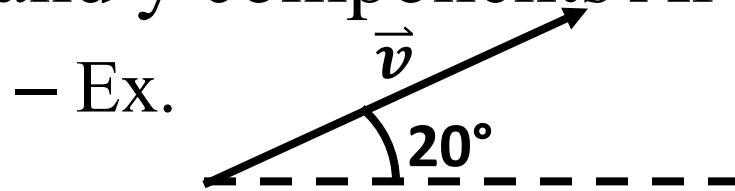
- Overview of course
- Define units/variables used in physics
- Define vectors and their notation
- Determine how to break vectors into components and combine separate vectors

- Classical Mechanics
 - Mechanics: The simple how (kinematics) and why (dynamics) things work
 - Motion, energy, vibrations
 - Classical
 - Not too fast ($v \ll c$)
 - Not too small ($d \gg$ atomic lengths)
- Accurately describes most everyday experiences
 - Path of a football
 - Orbits of planets around the sun
 - Vibrations of a guitar string

Math Skills

- **Algebra** – ability to solve for missing variable
 - Ex. $349 = (8)(9) + 3x^2$

- **Trigonometry** – ability to break vectors into x and y components AND recombine



$$v_x = v \cos 20^\circ$$

$$v_y = v \sin 20^\circ$$

$$v = \sqrt{v_x^2 + v_y^2}$$

- **Calculus** – Derivative & Integration

Problem Solving Strategy

0. Draw a picture if possible
1. Identify the givens
2. Identify what you need to solve for
3. Find equations/process that have the pieces from steps 1 and 2.*
4. Plug and chug

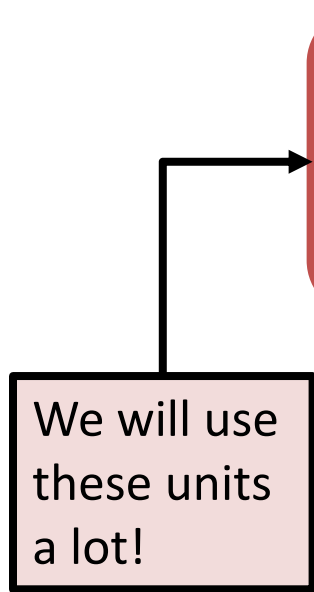
Units

- DON'T FORGET YOUR UNITS!!!
- All units plugged into problems must be base (**SI**) units
 - i.e. length (m), mass (kg), and time (s)
 - Ex. **3 mm** => **3 x 10⁻³ m** (same as 3E-3 m)
- You don't have to memorize this ☺

Prefix	Abbreviation	Exponential Multiplier	Meaning	Example using Length
giga	G	10 ⁹	1000000000	1 <u>gigameter</u> (Gm) = 1000000000m
mega	M	10 ⁶	1000000	1 megameter (Mm) = 1000000m
kilo	k	10 ³	1000	1 kilometer (km) = 1000m
hecto	h	10 ²	100	1 hectometer (hm) = 100m
deka	da	10 ¹	10	1 dekameter (dam) = 10m
			1	1 meter (m)
deci	d	10 ⁻¹	1/10	1 decimeter (dm) = -0.1m
centi	c	10 ⁻²	1/100	1 centimeter (cm) = 0.01m
milli	m	10 ⁻³	1/1000	1 millimeter (mm) = 0.001m
micro	μ	10 ⁻⁶	1/1000000	1 micrometer (μm) = 0.000001m
nano	n	10 ⁻⁹	1/1000000000	1 nanometer (nm) = 0.000000001m
pico	p	10 ⁻¹²	1/1000000000000	1 picometer (pm) = 0.000000000001m

Metric System

- A convenient system of units to use is the metric system (SI units)



We will use these units a lot!

ISQ Base Quantity	SI Base Unit
Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Electrical current	ampere (A)
Thermodynamic temperature	kelvin (K)
Amount of substance	mole (mol)
Luminous intensity	candela (cd)

Table 1.1 ISQ Base Quantities and Their SI Units

CQ #1

What is **38 km** in meters?

- a. $38 \times 10^3 \text{ m}$
- b. $38\text{E}3 \text{ m}$
- c. $38,000 \text{ m}$
- d. $3.8\text{E}4 \text{ m}$

Prefix	Abbreviation	Exponential Multiplier	Meaning	Example using Length
giga	G	10^9	1000000000	1 <u>gigameter</u> (Gm) = 1000000000m
mega	M	10^6	1000000	1 megameter (Mm) = 1000000m
kilo	k	10^3	1000	1 kilometer (km) = 1000m
hecto	h	10^2	100	1 hectometer (hm) = 100m
deka	da	10^1	10	1 dekameter (dam) = 10m
			1	1 meter (m)
deci	d	10^{-1}	1/10	1 decimeter (dm) = 0.1m
centi	c	10^{-2}	1/100	1 centimeter (cm) = 0.01m
milli	m	10^{-3}	1/1000	1 millimeter (mm) = 0.001m
micro	μ	10^{-6}	1/1000000	1 micrometer (μm) = 0.000001m
nano	n	10^{-9}	1/1000000000	1 nanometer (nm) = 0.000000001m
pico	p	10^{-12}	1/1000000000000	1 picometer (pm) = 0.000000000001m

ABCD card

How many **ms** are in a minute?

- a. 60 ms
- b. 6E4 ms (6×10^4 ms)
- c. 1E3 ms
- d. 10 ms

A

B

C

D

Conversion of Units

$$1 \text{ mi} = 1\,609 \text{ m} = 1.609 \text{ km}$$

$$1 \text{ ft} = 0.304\,8 \text{ m} = 30.48 \text{ cm}$$

$$1 \text{ m} = 39.37 \text{ in.} = 3.281 \text{ ft}$$

$$1 \text{ in.} = 0.025\,4 \text{ m} = 2.54 \text{ cm (exactly)}$$

Convert 15.0 in. to centimeters.

Conversion of Units

Convert 15.0 in. to centimeters.

$$1 \text{ in.} = 2.54 \text{ cm} \rightarrow \frac{2.54 \text{ cm}}{1 \text{ in.}}$$

$$15.0 \text{ in.} = (15.0 \cancel{\text{in.}}) \left(\frac{2.54 \text{ cm}}{1 \cancel{\text{in.}}} \right) = 38.1 \text{ cm}$$

Example (a) Is he speeding?

What if the driver were from outside the United States and were familiar with speeds measured in kilometers per hour? He saw 80.0 mph on the speedometer; what is the **speed** in km/h?

$$\left(85.0 \frac{\cancel{\text{mi}}}{\text{h}} \right) \left(\frac{1.609 \text{ km}}{1 \cancel{\text{mi}}} \right) = 137 \text{ km/h}$$



Example (b) Is he speeding?

On an interstate highway in rural Wyoming, a car is traveling at 33.0 m/s. Is the driver exceeding the 75.0 mi/h speed limit?

$$(33.0 \text{ m/s}) \left(\frac{1 \text{ mi}}{1609 \text{ m}} \right) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) \left(\frac{60 \text{ min}}{1 \text{ h}} \right) = 73.8 \text{ mi/h}$$

No, he is not speeding!

Problem solving #1 (Unit & Vectors)

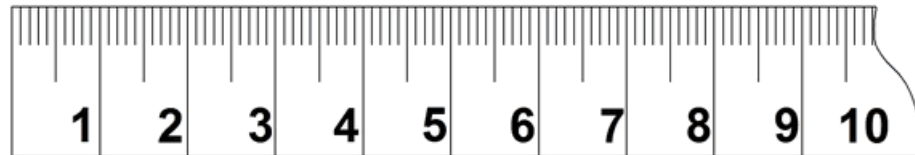
1) A car is traveling at a speed of 120 km/hr. (a) What is its speed in meters per second (m/s)?

Ans = 33.3 m/s

Significant Figures

Experimental numbers must be written in a way consistent with the precision to which they are known. In this context one speaks of **significant figures** or digits that have physical meaning.

1. All definite digits and the first doubtful digit are considered significant.
2. Leading zeros are **not** significant figures.
Example: $L = 2.31\text{cm}$ has 3 significant figures. For $L = 0.0231\text{m}$, the zeros serve to move the decimal point to the correct position. Leading zeros are not significant figures.
3. Trailing zeros are significant figures: they indicate the number's precision.
4. One significant figure should be used to report the uncertainty or occasionally two, especially if the second figure is a five.



Significant Figures

1500 g $\rightarrow$? significant figures

1.500×10^3 g $\rightarrow$ **4** significant figures

1.50×10^3 g $\rightarrow$ **3** significant figures

1.5×10^3 g $\rightarrow$ **2** significant figures

2.3×10^{-4} $\rightarrow$ **2** significant figures $\rightarrow$ 0.00023

2.30×10^{-4} $\rightarrow$ **3** significant figures $\rightarrow$ 0.000230

Significant Figures (add & subtract)

When numbers are added or subtracted, the number of decimal places in the result should equal the smallest number of decimal places of any term in the sum or difference.

$$23.2 + 5.174 = 28.374$$

→ 23.2 has one decimal place → sum = 28.4

Significant Figures (divide & multiply)

When multiplying several quantities, the number of significant figures in the final answer is the same as the number of significant figures in the quantity having the smallest number of significant figures. The same rule applies to division.

Example (c) Installing a carpet

A carpet is to be installed in a rectangular room whose length is measured to be 12.71 m and whose width is measured to be 3.46 m. Find the area of the room.

$$12.71 \text{ m} \times 3.46 \text{ m} = 43.9766 \text{ m}^2$$

$$3.46 \rightarrow 3 \text{ sig figs} \rightarrow A = 44.0 \text{ m}^2$$

Significant Figures (Scientific notation)

Find the area of a circular plate that has a radius of 6.0 cm.

$$A = \pi r^2 = \pi (6.0 \text{ cm})^2 = 1.1 \times 10^2 \text{ cm}^2$$

calculator answer: 113.0973355

~~113 cm~~

Scientific Notation!

Significant Figures (Rounding)

- **last digit dropped > 5:** increase last retained digit by 1:
 $1.35\text{\underline{6}} \rightarrow 1.36$
- **last digit dropped = 5:** increase last retained rounded to nearest even number: $1.33\text{\underline{5}} \rightarrow 1.34$ or $1.34\text{\underline{5}} \rightarrow 1.34$
- **last digit dropped < 5:** leave last retained as is: $1.34\text{\underline{3}} \rightarrow 1.34$

In our course, most of the numerical examples and end-of-chapter problems will yield answers having **three significant figures**.

Density

$$\rho \equiv \frac{m}{V}$$



Styrofoam



Lead

Convert the density of Lead 11.35 g/cm³ into kg/m³

Men's 50 m Freestyle swimming time



≡ MENU

START LIST RESULTS

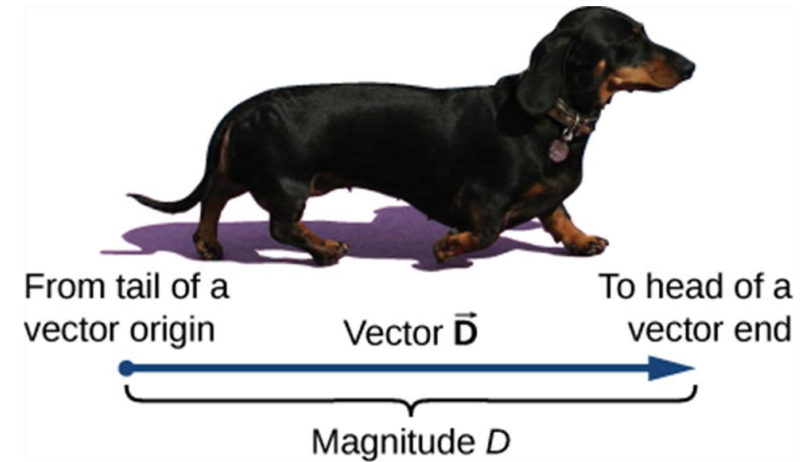
RANK	NAME	REACTION TIME	TIME
1	 MCEVOY Cameron AUS	0.56	21.25 
2	 PROUD Ben GBR	0.59	21.30 
3	 MANAUDOU Florent FRA	0.63	21.56 
4	 LIENDO Joshua CAN	0.64	21.58 
5	 GKOLOMEEV Kristian GRE	0.63	21.59 
6	 DRESSEL Caeleb USA	0.63	21.61 
7	 DEPLANO Leonardo ITA	0.66	21.62 
8	 CROOKS Jordan CAY	0.63	21.64 

Ch 1: Summary

- DON'T FORGET YOUR **UNITS!!!!**
- Make sure your numbers are in **base** units
- **Conversion** of a number from one unit into another
- **Significant figures**
- **Rule** for adding/subtracting and multiplying/dividing significant figures
- **Scientific** notation
- **Rounding** a number

Vectors

- Scalar vs Vector Quantities
- Coordinate system and component of vectors
- Unit Vector
- Vector addition and subtraction
- Vector operation using components



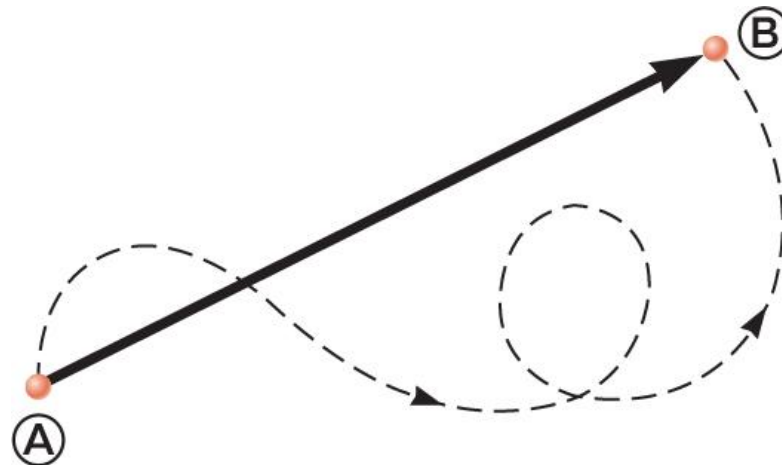
Helpful Video: Scalars, Vectors, and Vector Operations:

<https://www.youtube.com/watch?v=sXKiAKn0WCM&t=181s>

Vectors and Scalars

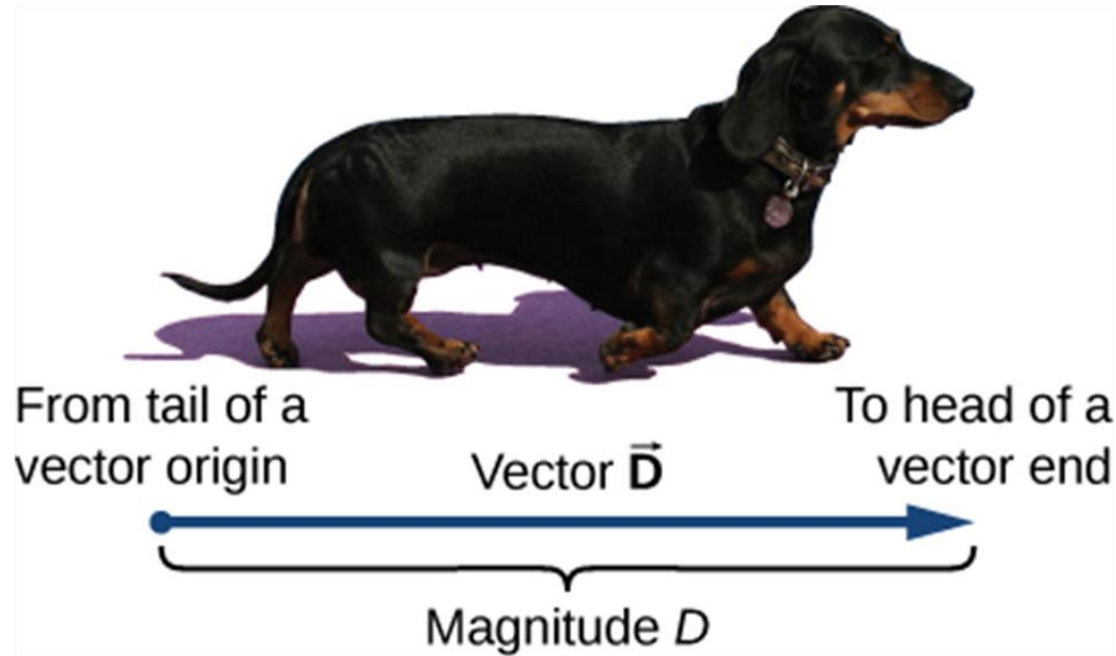
A **scalar quantity** is completely specified by a single value with an appropriate unit and has ***no direction***.

A **vector quantity** is completely specified by a number with an appropriate unit (the magnitude of the vector) plus ***a direction***.



Displacement
Vector

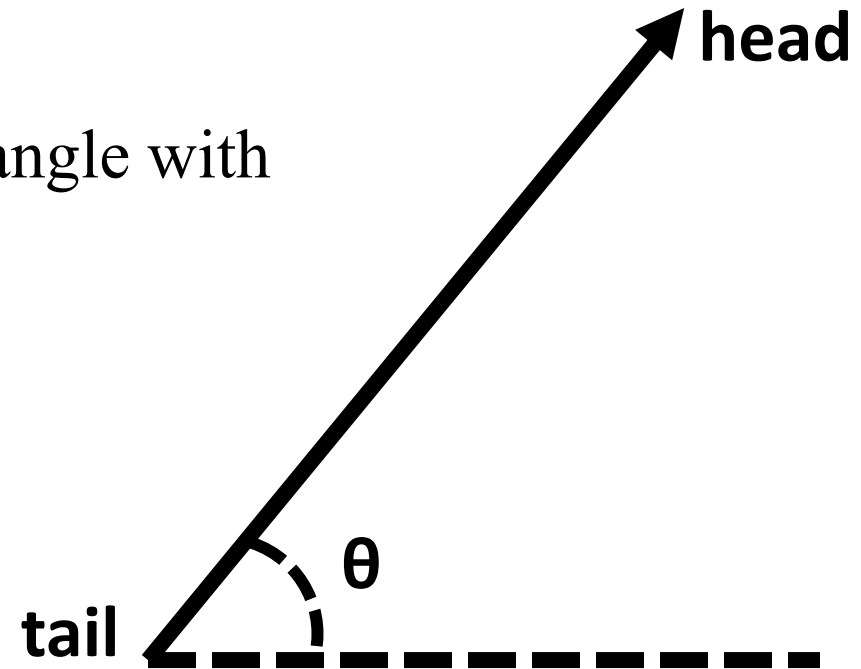
Vectors Magnitude + Direction



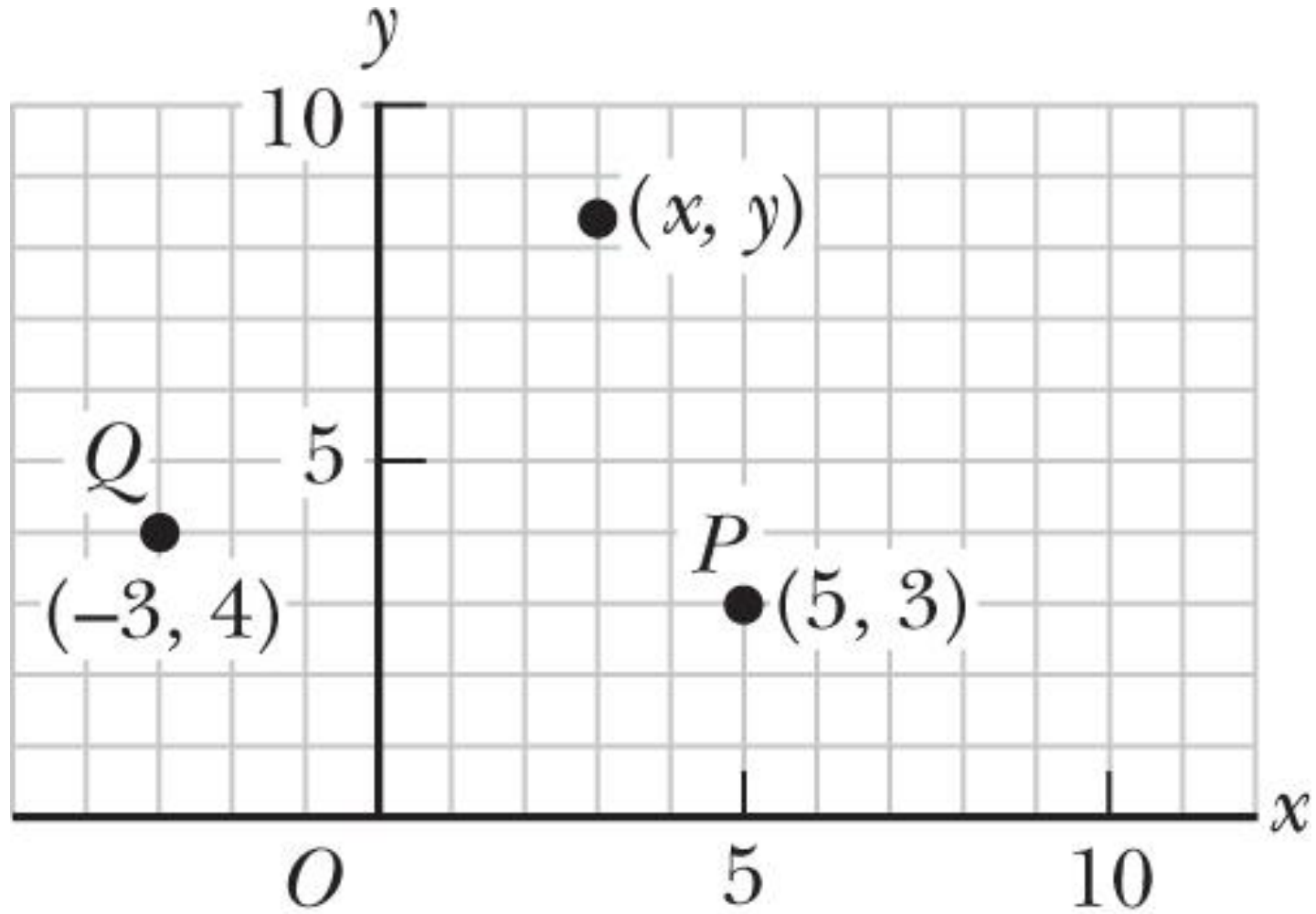
- We draw a vector from the initial point or origin (called the “tail” of a vector) to the end or terminal point (called the “head” of a vector), marked by an arrowhead. Magnitude is the length of a vector and is always a positive scalar quantity.

Vectors

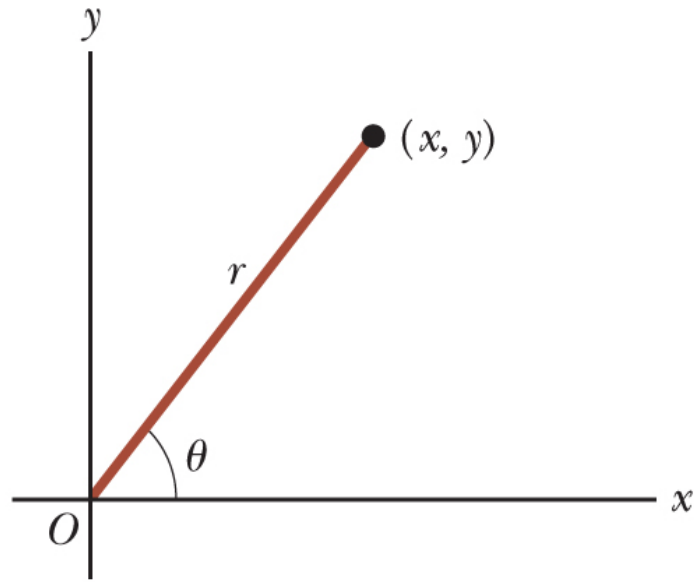
- Vectors have magnitude and direction
- Arrows are used to represent vectors
 - Length is measured tail to head and is used to represent magnitude
 - Direction is often given by angle with respect to x-axis



Cartesian Coordinate System



Polar Coordinate System



a

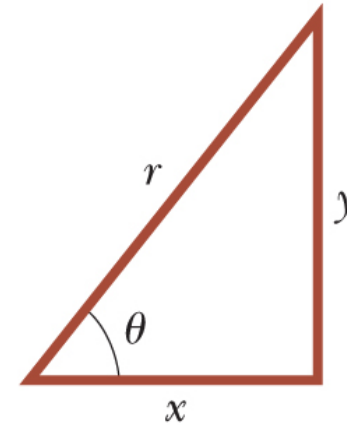
$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x}$$



b

$$\tan \theta = \frac{y}{x}$$

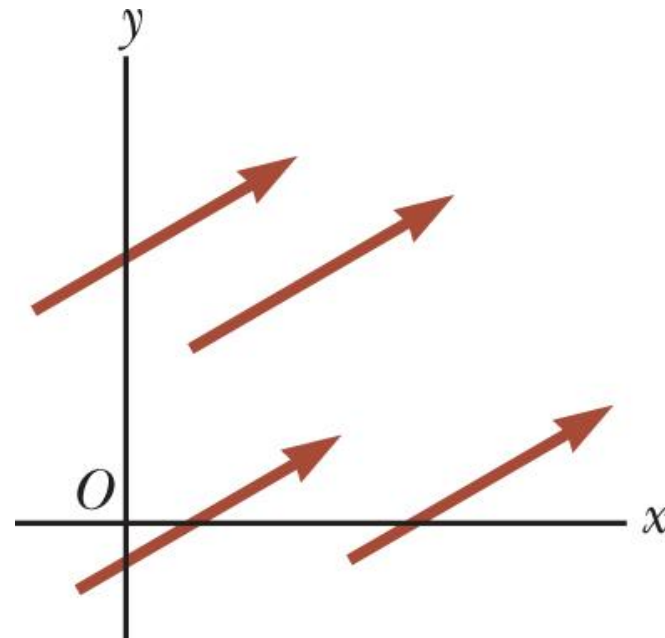
$$r = \sqrt{x^2 + y^2}$$

Basic Vector Arithmetic

$$\vec{A} = \vec{B} \text{ only if } A = B$$

Both point in the same direction along parallel lines

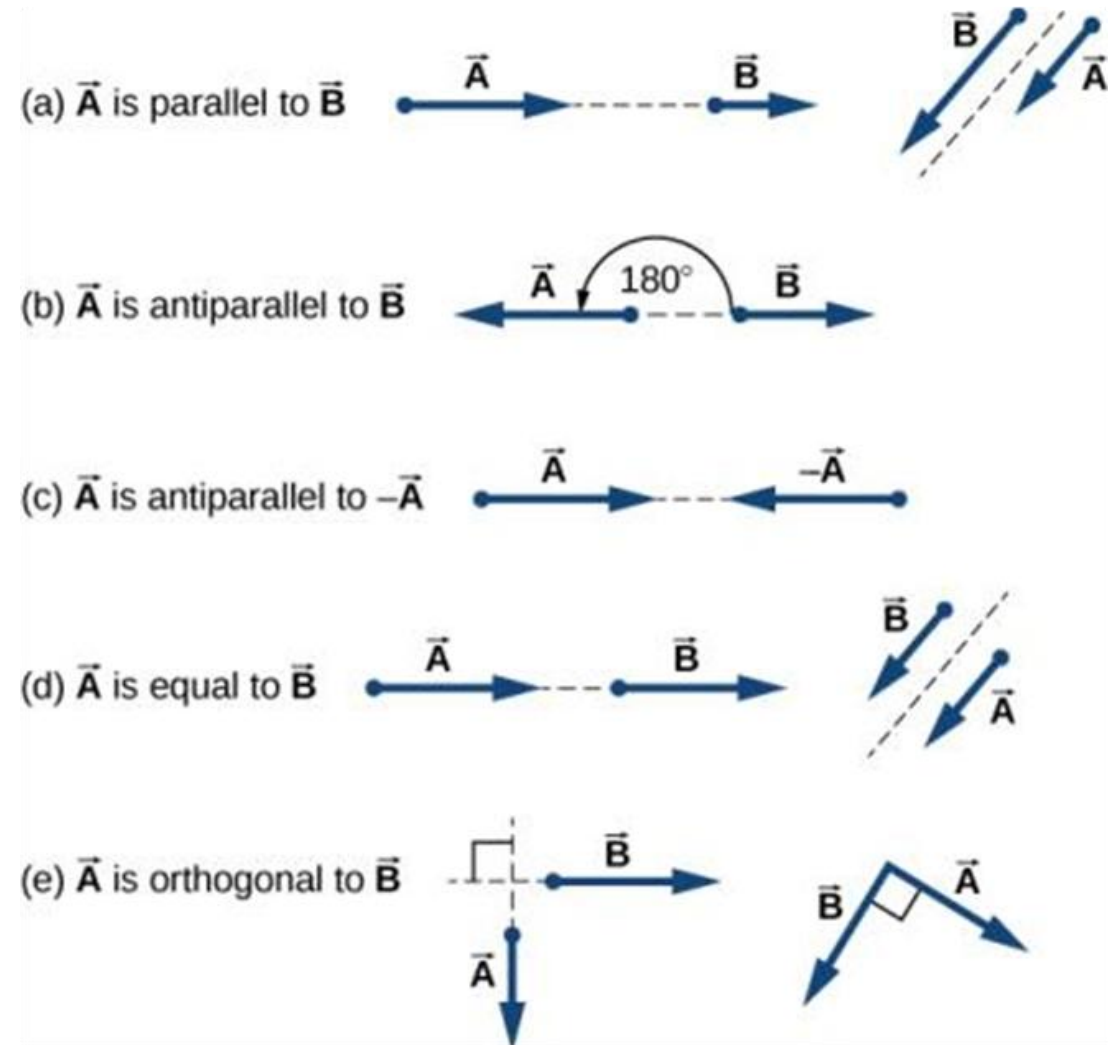
(magnitude + direction)



Comparing two Vectors

- Various relations between two vectors $\vec{A}$ and $\vec{B}$:

- (a) $\vec{A} \neq \vec{B}$ because $A \neq B$.
- (b) $\vec{A} \neq \vec{B}$ because they are not parallel and $A \neq B$.
- (c) $\vec{A} \neq -\vec{A}$ because they have different directions (despite identical magnitude).
- (d) $\vec{A} = \vec{B}$ because they are parallel and have identical magnitudes $A = B$.
- (e) $\vec{A} \neq \vec{B}$ because they have different directions (are not parallel); here, their directions differ by 90° —meaning, they are orthogonal.



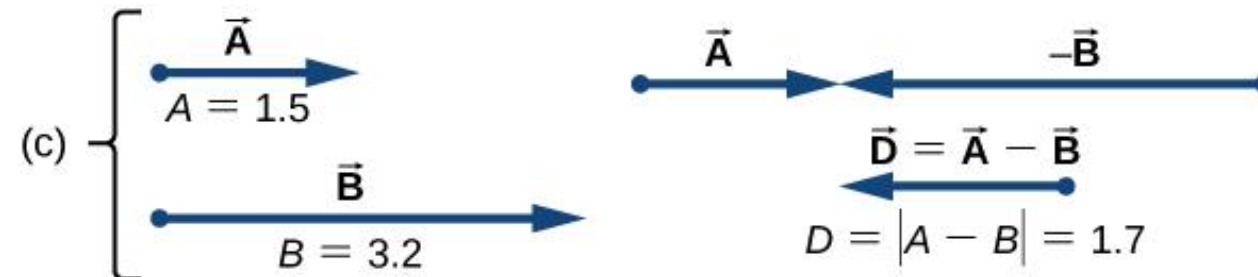
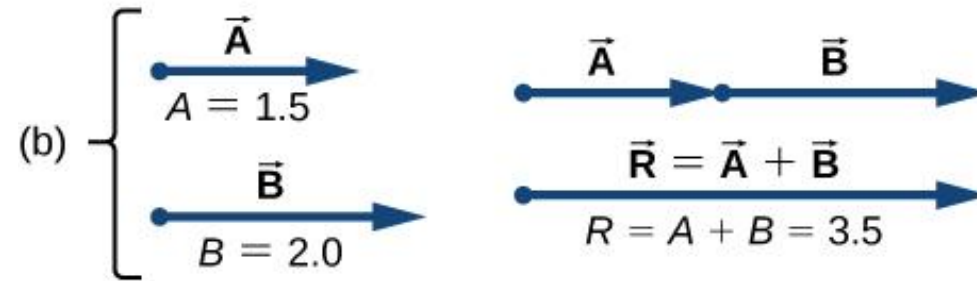
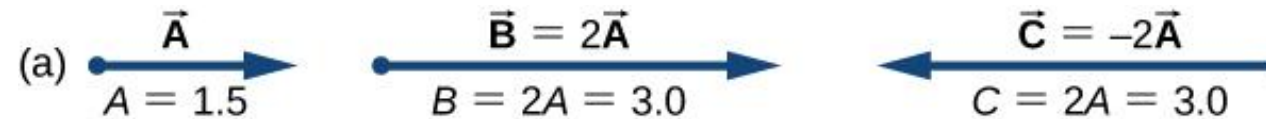
Vector Algebra in 1D

- Algebra of vectors in one dimension:

(a) Multiplication by a scalar.

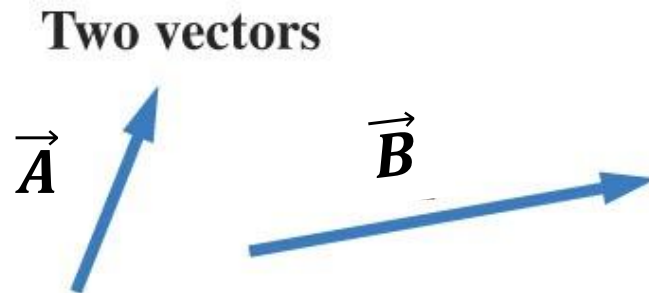
(b) Addition of two vectors ($\vec{R}$ is called the *resultant* of vectors $\vec{A}$ and $\vec{B}$).

(c) Subtraction of two vectors ($\vec{D}$ is the difference of vectors $\vec{A}$ and $\vec{B}$).



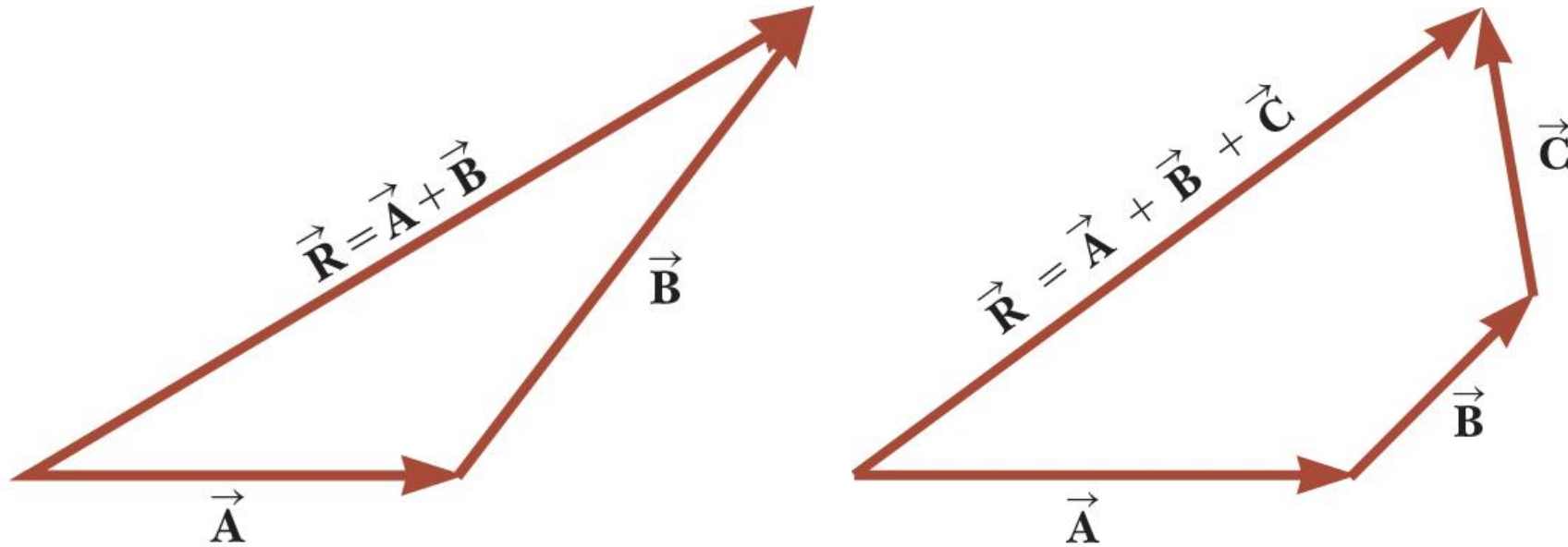
Vector Addition visually (graphically) in 2D

- You can add vectors visually and numerically
- To add vectors, we draw the vectors tip(head)-to-tail
 1. Draw the first vector
 2. Start the next vectors **tail at the tip** of the first vector
 3. Repeat till all vectors are added
- Ex.



Adding vectors visually (graphically)

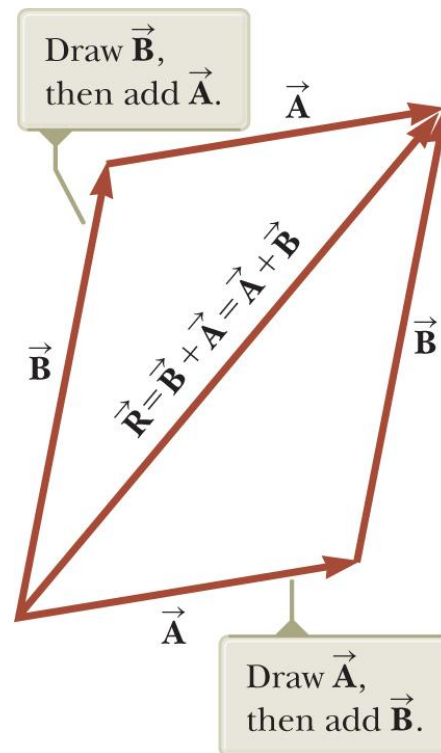
Head-to-tail method



Vectors Addition in 2D

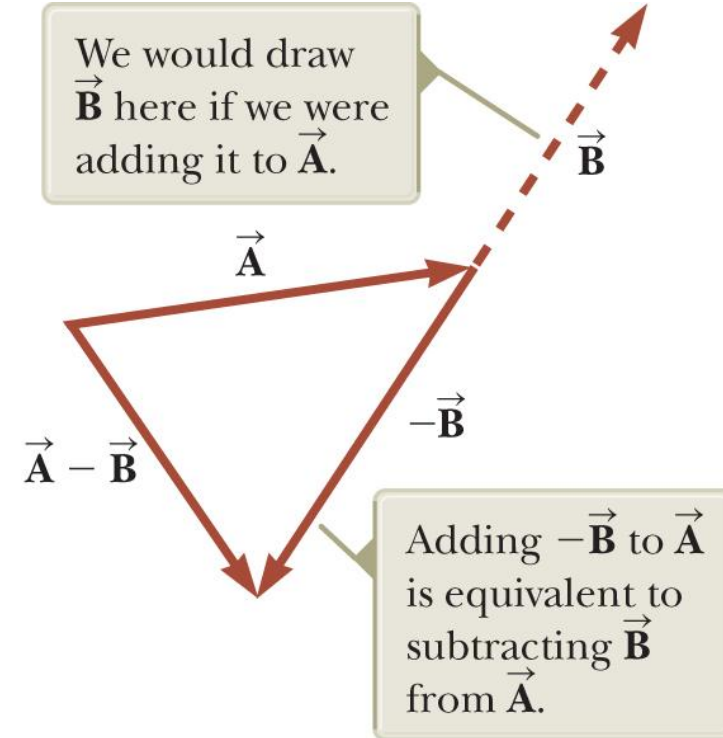
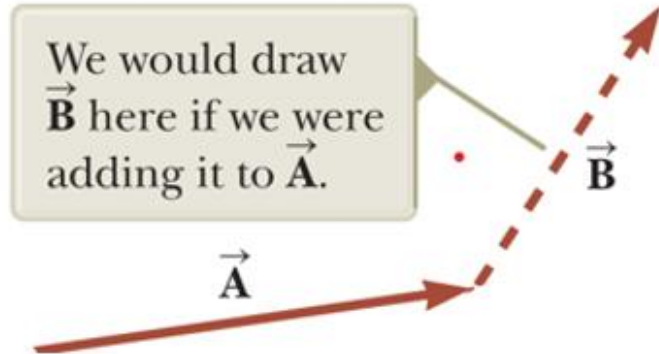
$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \text{ (commutative law of addition)}$$

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \text{ (associative law of addition)}$$



Vectors **Subtraction** in 2D

$$\vec{\mathbf{A}} - \vec{\mathbf{B}} = \vec{\mathbf{A}} + (-\vec{\mathbf{B}})$$

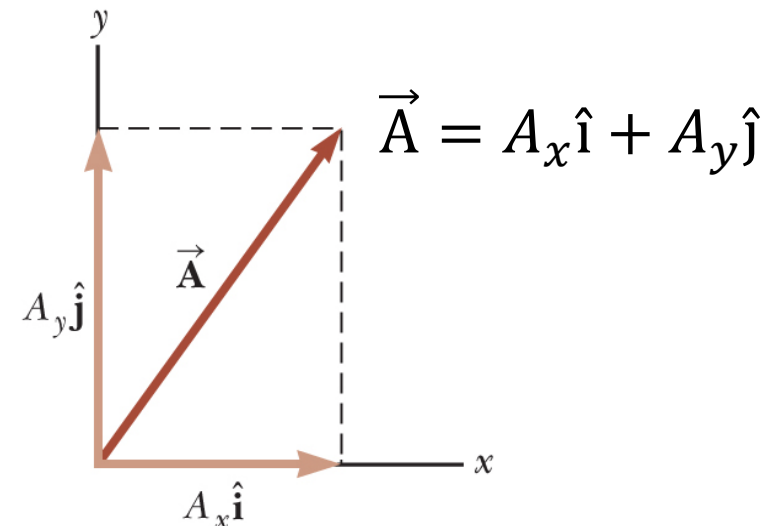
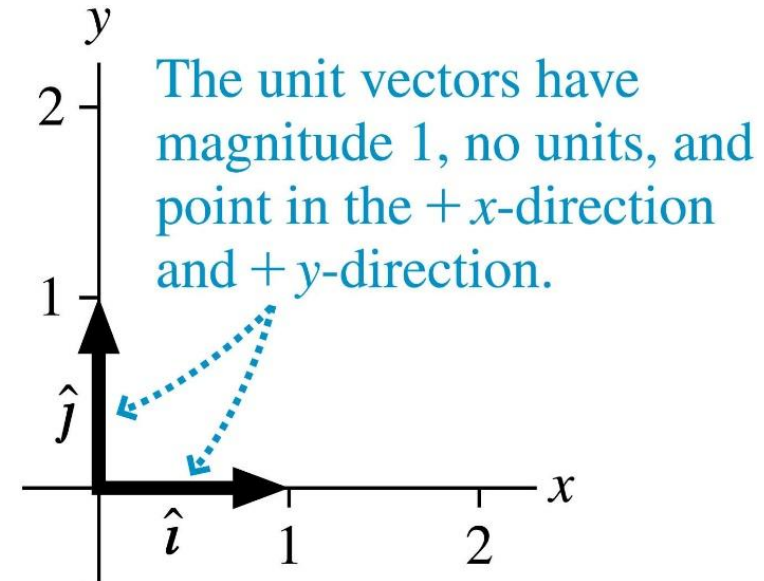


Unit Vector

- Each vector in the figure to the right has a magnitude of 1, no units, and is parallel to a coordinate axis.
- A vector with these properties is called a **unit vector**.
- These unit vectors have the special symbols:

$\hat{i} \equiv (1, \text{positive } x\text{-direction})$

$\hat{j} \equiv (1, \text{positive } y\text{-direction})$

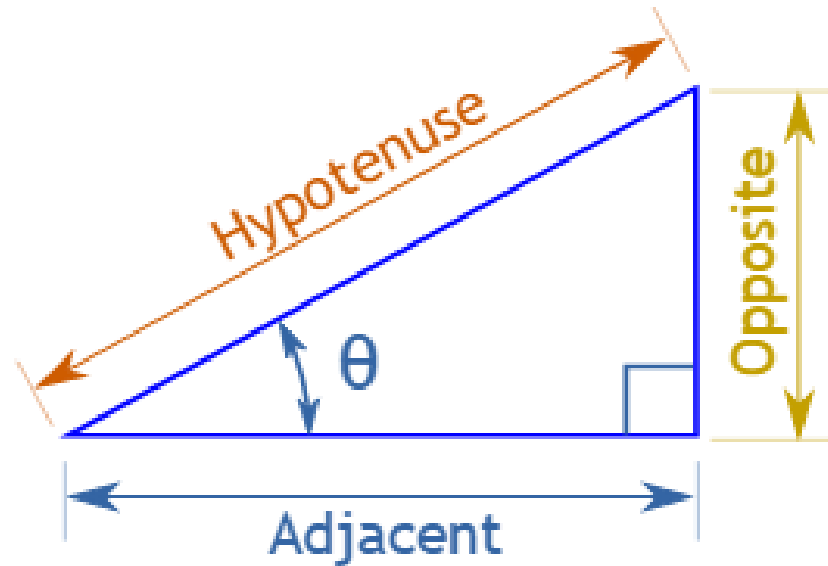


Trigonometry

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

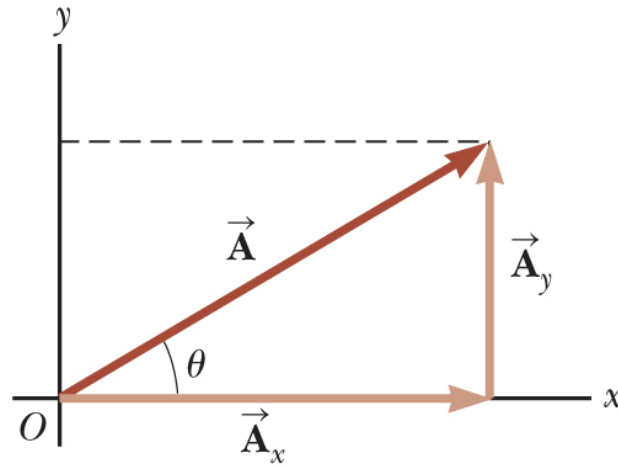
$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

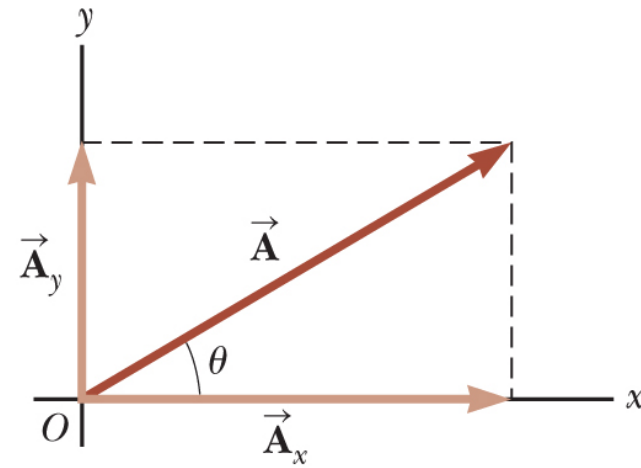


SOHCAHTOA

Component of a Vector



a



b

SOHCAHTOA

$$\cos \theta = \frac{A_x}{A} \Rightarrow A_x = A \cos \theta$$
$$\sin \theta = \frac{A_y}{A} \Rightarrow A_y = A \sin \theta$$

Pythagorean Theorem

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

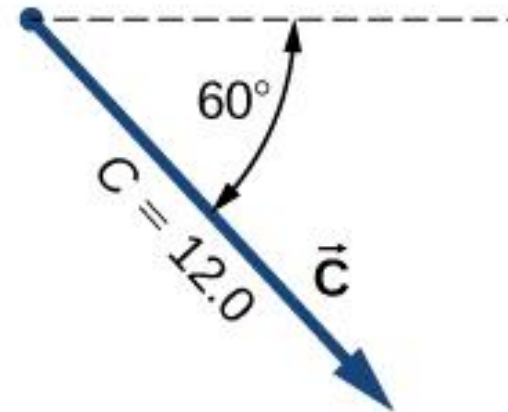
Problem solving

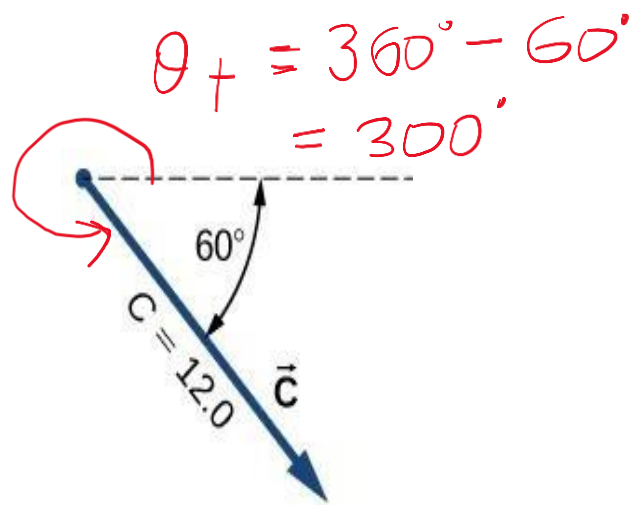
Write down the following vector in vector component form:

Answer:

$$C_x = 6.00$$

$$C_y = -10.4$$





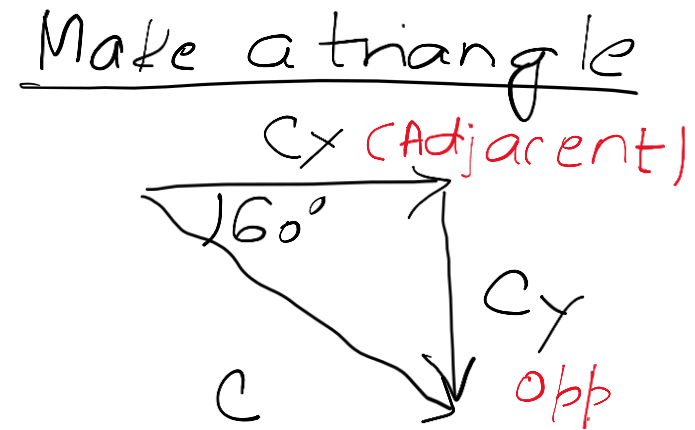
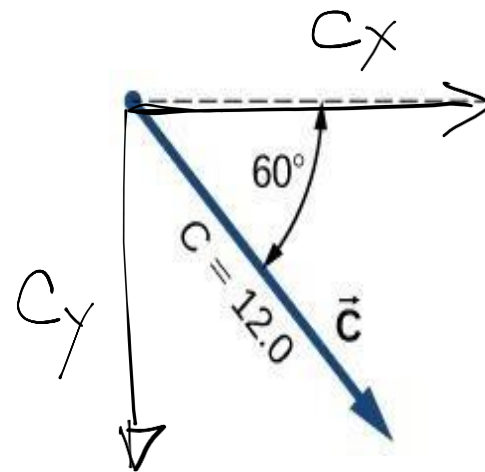
If you use positive angle, angle will take care of direction of the vector components.

$$a) C_x = C \cos \theta_+ = 12.0 \cos(300^\circ)$$

$$\boxed{C_x = 6.00}$$

$$b) C_y = C \sin \theta_+ = 12.0 \sin(300^\circ)$$

$$\boxed{C_y = -10.4}$$



Use SOHCAHTOA

$$\sin 60^\circ = \frac{C_y}{c} \Rightarrow C_y = C \sin 60^\circ$$

$$\boxed{C_y = -10.4}$$

You have to keep track of direction of vector component and ADD a negative sign!

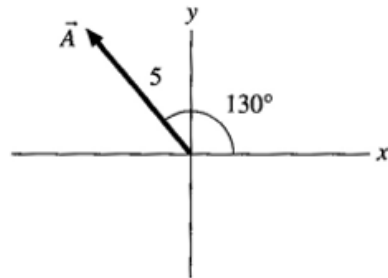
$$\cos 60^\circ = \frac{C_x}{c} \Rightarrow C_x = C \cos 60^\circ$$

$$\boxed{C_x = 6.00}$$

Component of a Vector

Determine the numerical values of the x- and y-components of each vector:

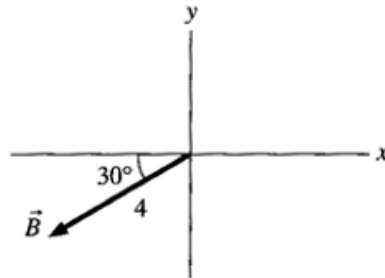
13.



$$A_x = \underline{\hspace{2cm}}$$

$$A_y = \underline{\hspace{2cm}}$$

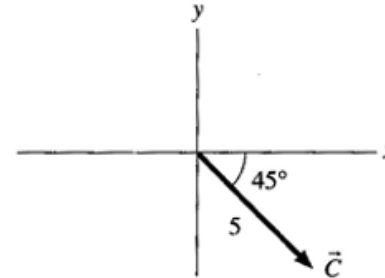
14.



$$B_x = \underline{\hspace{2cm}}$$

$$B_y = \underline{\hspace{2cm}}$$

15.

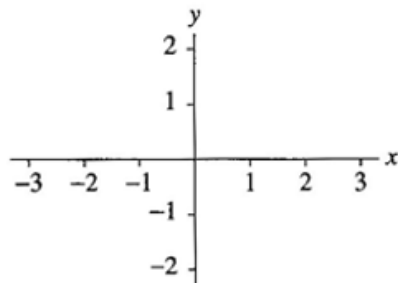


$$C_x = \underline{\hspace{2cm}}$$

$$C_y = \underline{\hspace{2cm}}$$

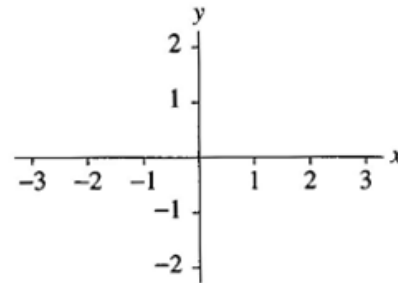
Draw and label the vector with these components. Then determine the magnitude of the vector.

16. $A_x = 3, A_y = -2$



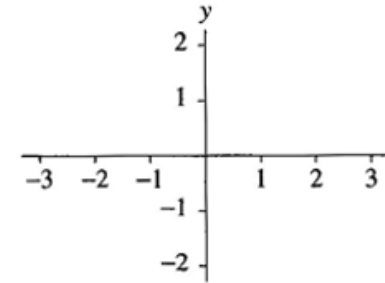
$$A = \underline{\hspace{2cm}}$$

17. $B_x = -2, B_y = 2$



$$B = \underline{\hspace{2cm}}$$

18. $C_x = 0, C_y = -2$

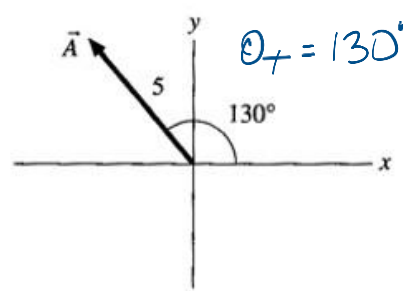


$$C = \underline{\hspace{2cm}}$$

Component of a Vector

Determine the numerical values of the x- and y-components of each vector:

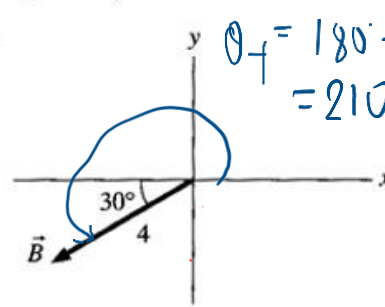
13.



$$A_x = 5 \cos 130^\circ$$

$$A_y = 5 \sin 130^\circ$$

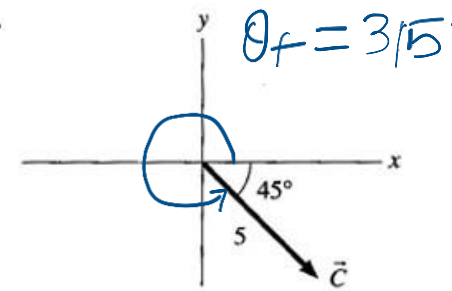
14.



$$B_x = 4 \cos 210^\circ$$

$$B_y = 4 \sin 210^\circ$$

15.

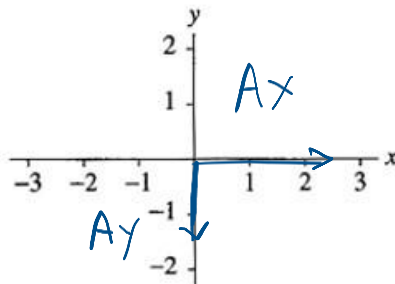


$$C_x = 5 \cos 315^\circ$$

$$C_y = 5 \sin 315^\circ$$

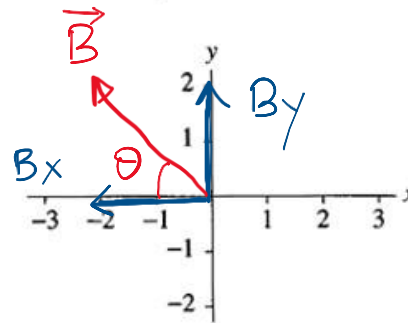
Draw and label the vector with these components. Then determine the magnitude of the vector.

16. $A_x = 3, A_y = -2$



$$A = \sqrt{3^2 + (-2)^2}$$

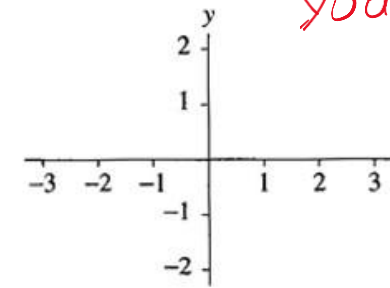
17. $B_x = -2, B_y = 2$



$$B = \sqrt{(-2)^2 + (2)^2}$$

$$\theta = \tan^{-1}(B_y/B_x)$$

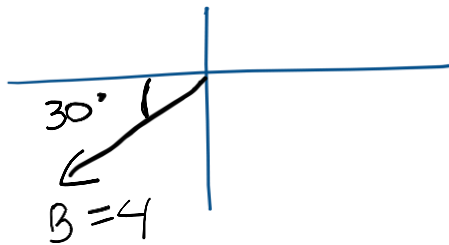
18. $C_x = 0, C_y = -2$



$$C = \sqrt{(0)^2 + (-2)^2}$$

Make sure you add 180° to the answer you get from calculator as $\vec{B}$ lies on the II quadrant!

14)



a) Take $\theta = 180^\circ + 30^\circ = 210^\circ$ (previous slide)

b) Use SOHCAHTOA

$$\cos 30^\circ = -\frac{B_x}{B} \quad (B_x \text{ is pointing to the left})$$

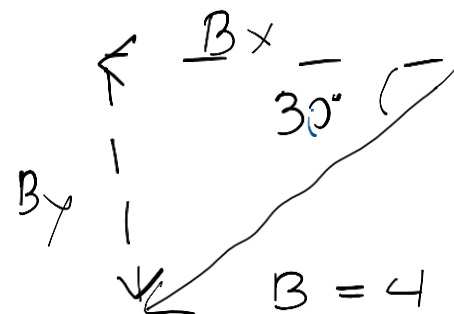
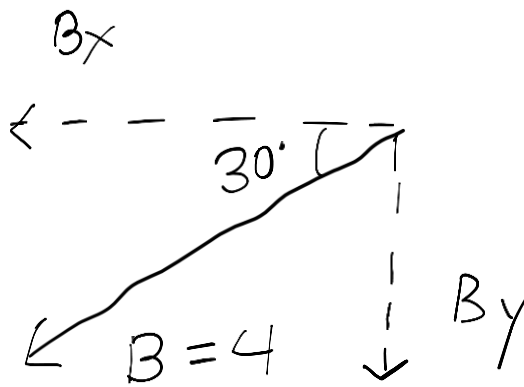
$$B_x = -B \cos 30^\circ$$

$$\boxed{B_x = -3.46}$$

$$\sin 30^\circ = -\frac{B_y}{B} \quad (B_y \text{ pointing downward})$$

$$B_y = -B \sin 30^\circ$$

$$\boxed{B_y = -2.00}$$



$$17) B_x = -2, B_y = 2$$

$$|\vec{B}| = \sqrt{(B_x)^2 + (B_y)^2}$$

$$= \sqrt{(-2)^2 + (2)^2} = \sqrt{4+4}$$

$$|\vec{B}| = 2.83$$

To find the direction of $\vec{B}$

$$\theta = \tan^{-1} \left(\frac{B_y}{B_x} \right)$$

$$= \tan^{-1} \left(-\frac{2}{2} \right)$$

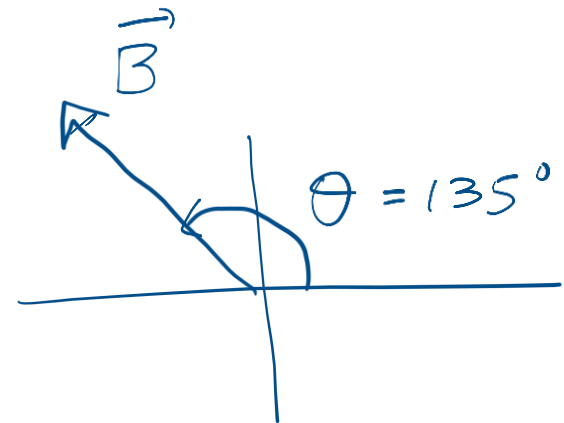
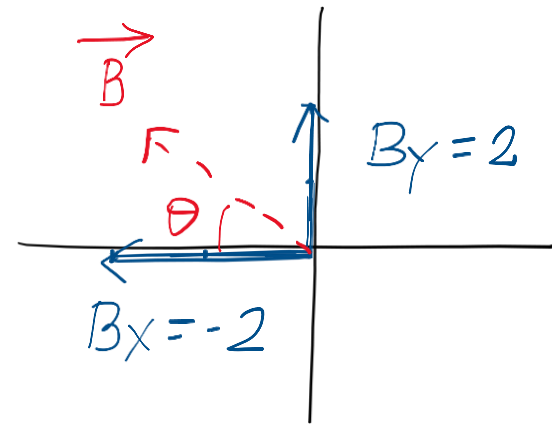
$$\theta = -45^\circ$$

For the correct value

$$\theta = -45^\circ + 180^\circ \leftarrow$$

$$\theta = 135^\circ$$

Add 180° whenever the vector is in II or III quadrant!!!



CQ #2

Which of the following shows the vector: $-15.79\hat{i} + 12.04\hat{j}$

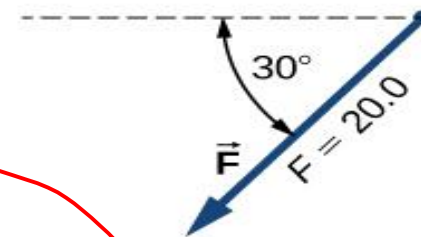
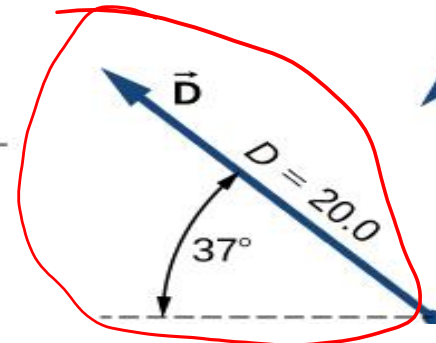
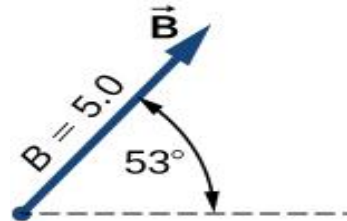
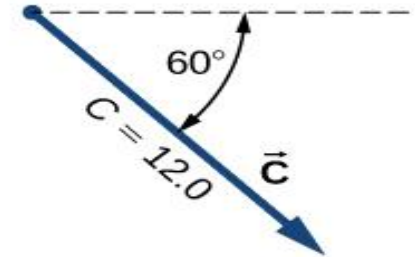
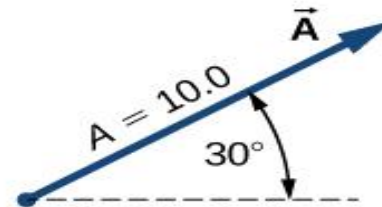
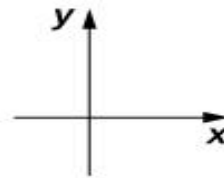
a) $\vec{A}$

b) $\vec{B}$

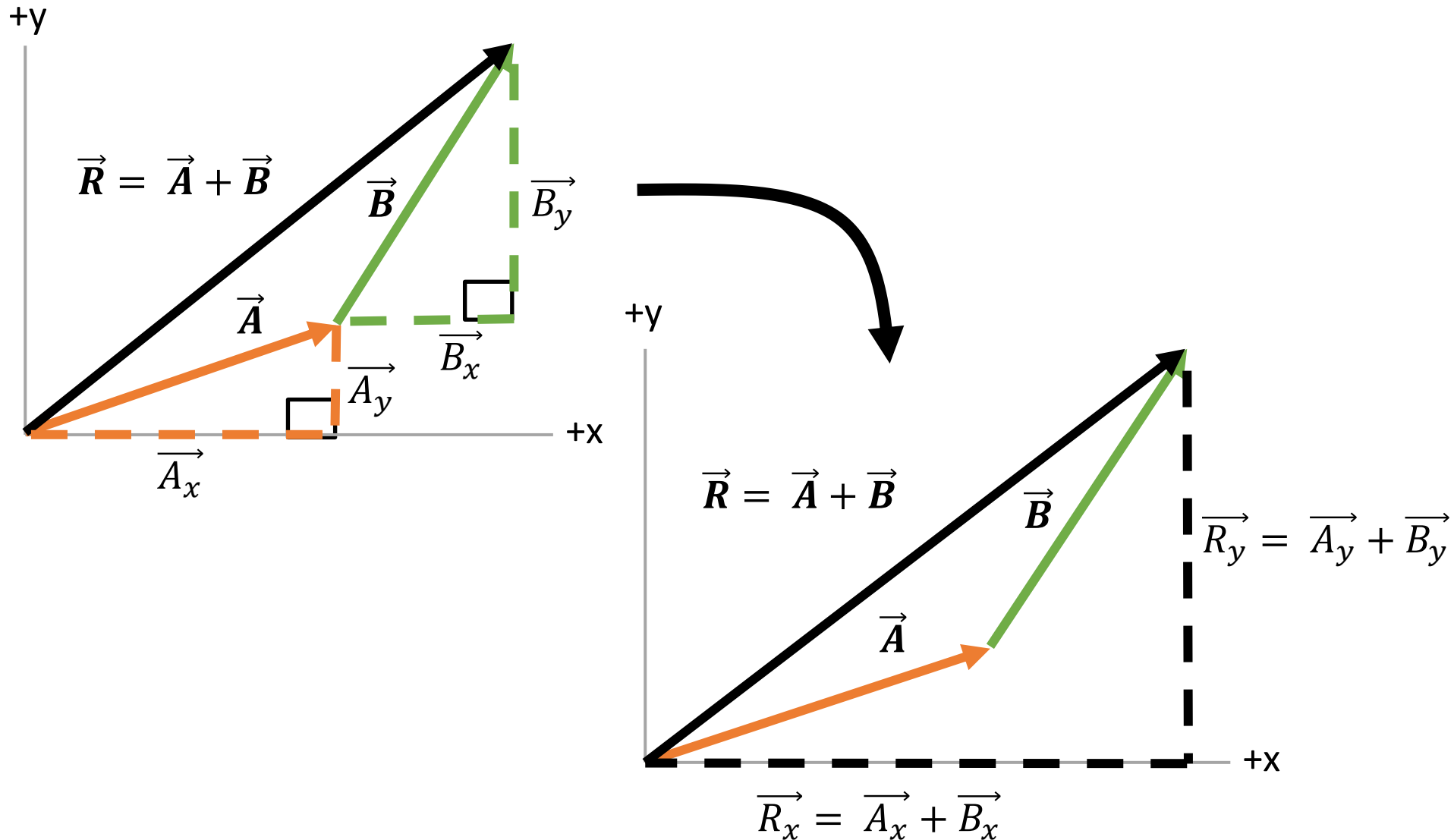
c) $\vec{C}$

d) $\vec{D}$

e) $\vec{F}$



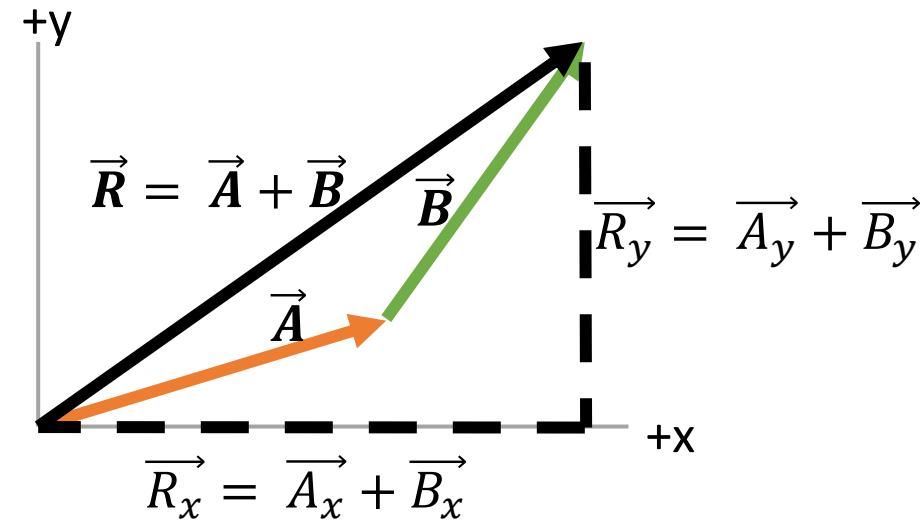
Vector addition using **components**



Vector addition using **components**

$$\begin{aligned}\vec{R} &= \vec{A} + \vec{B} = (A_x\hat{i} + A_y\hat{j}) + (B_x\hat{i} + B_y\hat{j}) \\ &= (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}\end{aligned}$$

$$\begin{aligned}\vec{R} &= R_x\hat{i} + R_y\hat{j} \\ \Rightarrow R_x &= A_x + B_x \text{ and } R_y = A_y + B_y\end{aligned}$$



Magnitude

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(A_x + B_x)^2 + (A_y + B_y)^2}$$

Direction

$$\tan \theta = \frac{R_y}{R_x}$$

Example 1

Find the sum of two vectors $\vec{A}$ and $\vec{B}$ lying in the xy plane and given by,

$$\vec{A} = (2.0\hat{i} + 2.0\hat{j}) \text{ and } \vec{B} = (2.0\hat{i} - 4.0\hat{j})$$

$$\vec{A} = A_x\hat{i} + A_y\hat{j} \qquad \vec{B} = B_x\hat{i} + B_y\hat{j}$$

$$A_x = 2.0, A_y = 2.0$$

$$B_x = 2.0, B_y = -4.0$$

Example 1

$$\vec{R} = R_x \hat{i} + R_y \hat{j}$$

$$\begin{aligned}\vec{R} &= (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} \\ &= (2.0 + 2.0)\hat{i} + (2.0 - 4.0)\hat{j} \\ &= 4.0\hat{i} - 2.0\hat{j}\end{aligned}$$

Magnitude

$$\begin{aligned}R &= \sqrt{R_x^2 + R_y^2} = \sqrt{(4.0)^2 + (-2.0)^2} \\ &= \sqrt{20} = \boxed{4.5}\end{aligned}$$

Direction

$$\tan \theta = \frac{R_y}{R_x} = \frac{-2.0}{4.0} = -0.50 \rightarrow \boxed{\theta = -27.0^\circ \text{ OR } 333^\circ}$$

ABCD card

What is $\vec{A} + \vec{B}$ if $\vec{A} = +3\hat{i} + 2\hat{j}$ and
 $\vec{B} = -5\hat{i} + 4\hat{j}$

a) $8\hat{i} + 6\hat{j}$

b) $-2\hat{i} + 6\hat{j}$

c) $2\hat{i} - 2\hat{j}$

d) $-2\hat{i} + 4\hat{j}$

$\checkmark$ $R = \underbrace{2\hat{i}}_{R_x} + \underbrace{6\hat{j}}_{R_y}$

Problem solving

Find the **resultant** of vector *along x and y – axis if*

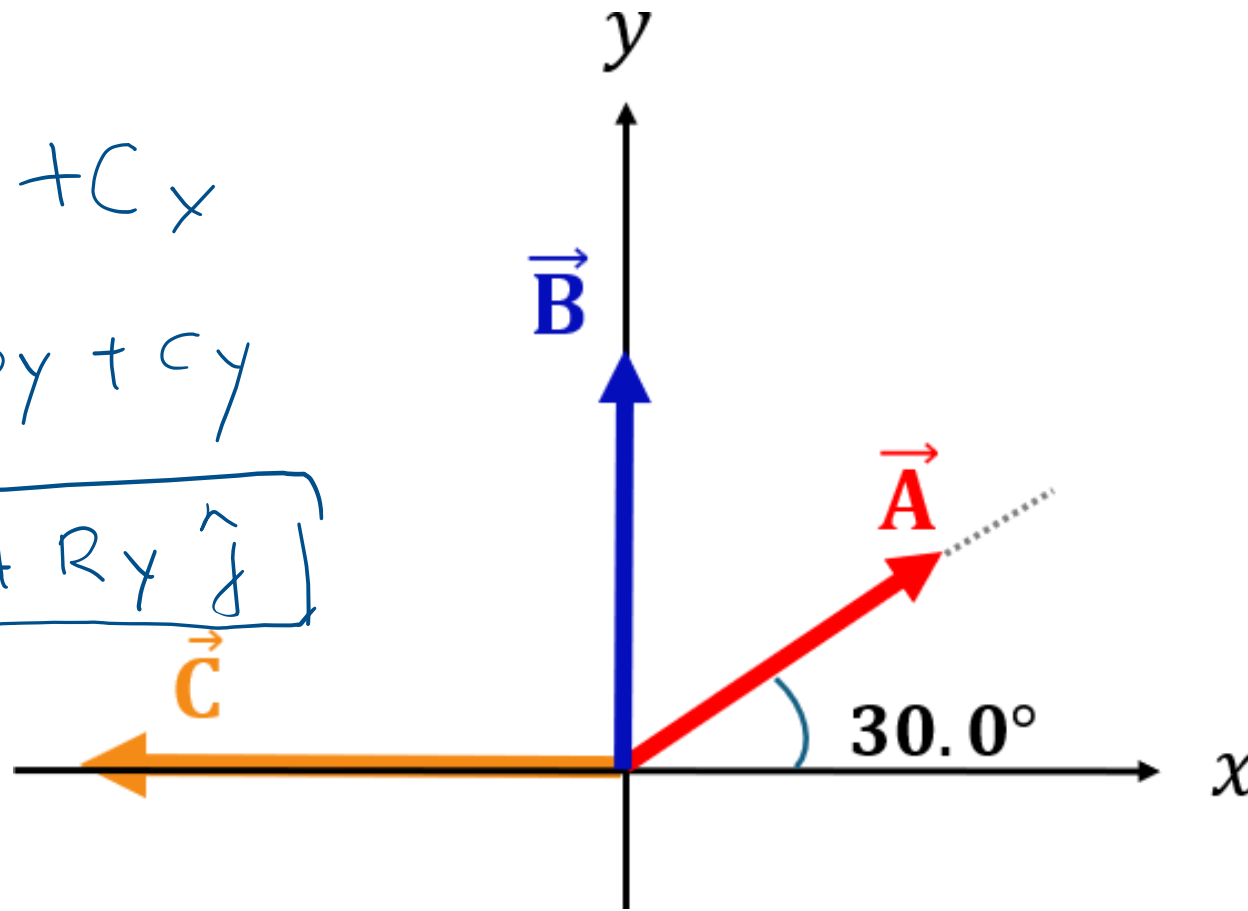
$|\vec{A}| = 5.00$ units, and $|\vec{B}| = 7.00$ units, and $|\vec{C}| = 8.50$ units.

$$R_x = A_x + B_x + C_x$$

$$R_y = A_y + B_y + C_y$$

$$\boxed{\vec{R} = R_x \hat{i} + R_y \hat{j}}$$

$$|\vec{R}| = ?$$
$$\theta = ???$$



$$A_x = A \cos 30^\circ$$

$$A_y = A \sin 30^\circ$$

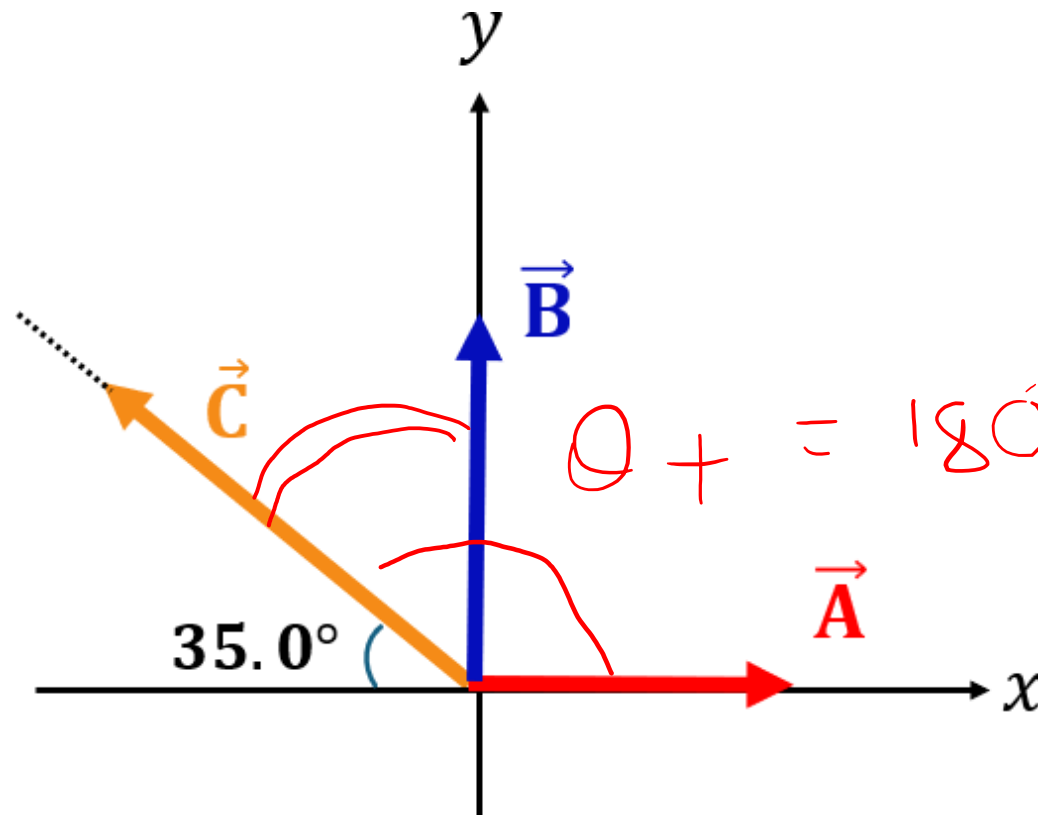
$$B_x = 0, B_y = 7.00$$

$$C_x = -8.50, C_y = 0$$

2

Problem solving

Find the **resultant** of vector *along x and y – axis* when $|\vec{A}| = 5.00 \text{ units}$, and $|\vec{B}| = 7.00 \text{ units}$, and $|\vec{C}| = 8.50 \text{ units}$.



$$\theta + = 180^\circ - 35^\circ = 145^\circ$$

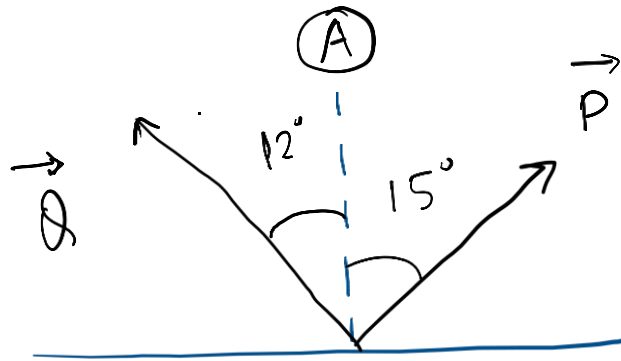
$$C_x = ?$$

$$C_y = ?$$

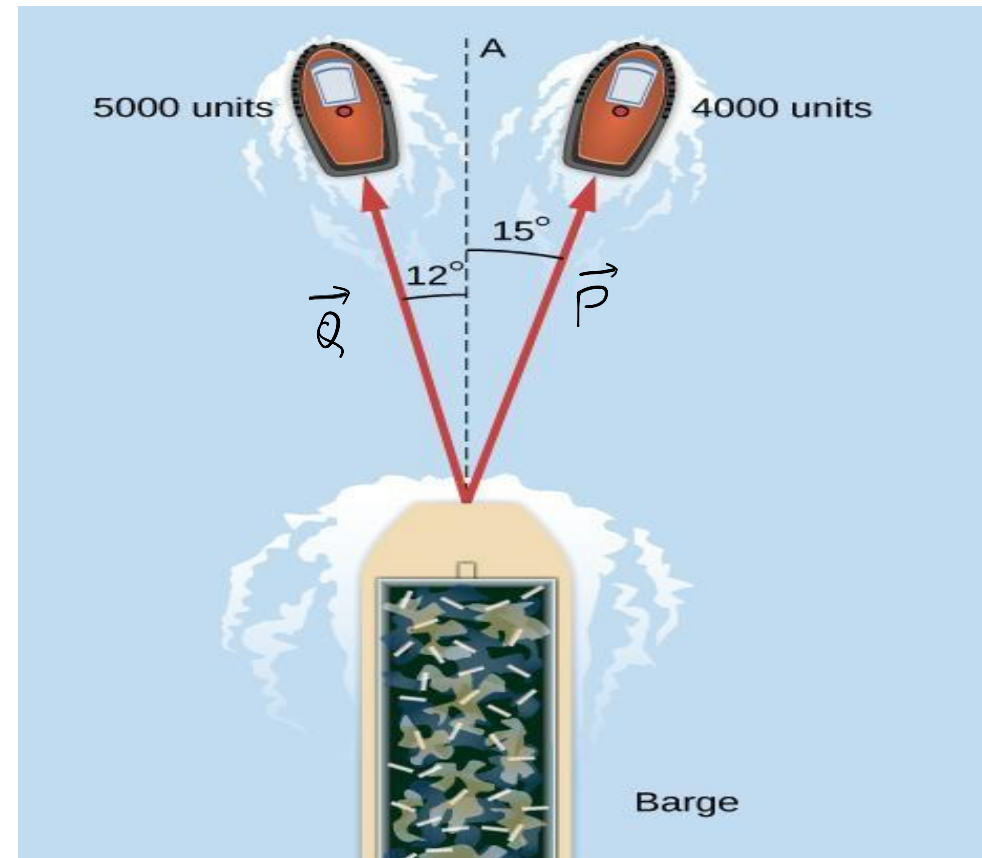
Problem solving

What is the net force in the following image and in which direction?

**Answer: 8754 units of force
approximately along A**



$$\vec{R} = \underbrace{(P_x + Q_x)}_{R_x} \hat{i} + \underbrace{(P_y + Q_y)}_{R_y} \hat{j}$$



To find component of vector $\vec{P}$,

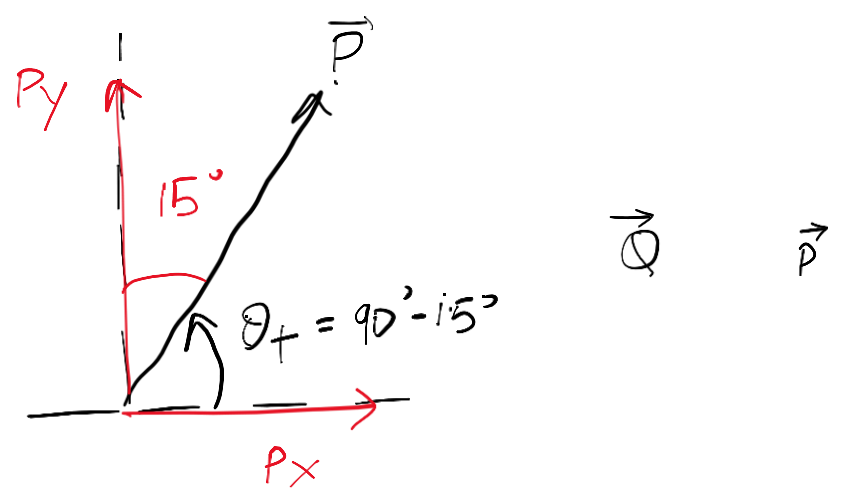
a) Use positive angle (θ_+) = $90^\circ - 15^\circ$

$$\theta_+ = 75^\circ$$

$$P_x = 4000 \cos 75^\circ \quad | \quad P_y = 4000 \sin 75^\circ$$

$$\boxed{P_x = 1035.3}$$

$$\boxed{P_y = 3863.7}$$



b) Use SDHCAHTOA

$$\cos 15^\circ = \frac{A}{H} = \frac{P_y}{\vec{P}}$$

$$P_y = \vec{P} \cos 15^\circ$$

$$= 4000 \cos 15^\circ$$

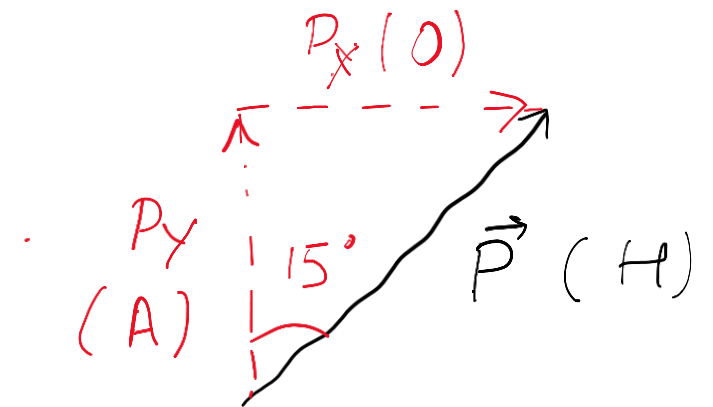
$$\boxed{P_y = 3863.7}$$

$$\sin 15^\circ = \frac{O}{H} = \frac{P_x}{\vec{P}}$$

$$P_x = \vec{P} \sin 15^\circ$$

$$= 4000 \sin 15^\circ$$

$$\boxed{P_x = 1035.3}$$



Similarly find the components of vector $\vec{Q}$ by either method

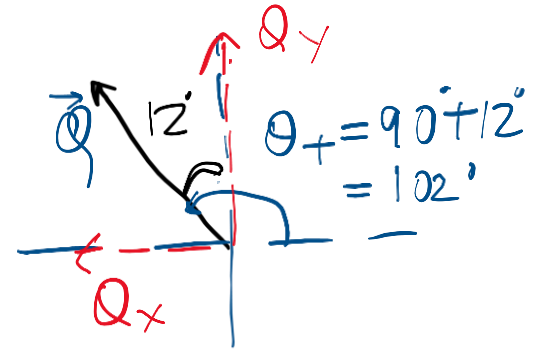
a) Use positive angle method

$$Q_x = Q \cos 102^\circ \\ = 5000 \cos 102^\circ$$

$$\boxed{Q_x = -1039.6}$$

$$Q_y = Q \sin 102^\circ \\ = 5000 \sin 102^\circ$$

$$\boxed{Q_y = 4890.7}$$



b) Use SOHCAHTOA

$$\cos 12^\circ = \frac{A}{H}$$

$$\cos 12^\circ = \frac{Q_y}{Q}$$

$$Q_y = 5000 \cos 12^\circ$$

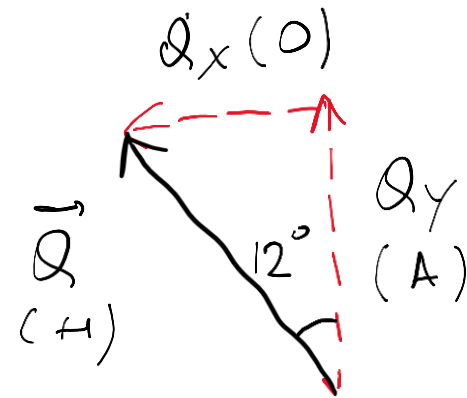
$$\boxed{Q_y = 4890.7}$$

$$\sin 12^\circ = \frac{O}{H}$$

$$\sin 12^\circ = \frac{-Q_x}{Q}$$

$$Q_x = -5000 \sin 12^\circ$$

$$\boxed{Q_x = -1039.6}$$



$$\begin{aligned}
 \vec{R} &= (P_x + Q_x) \hat{i} + (P_y + Q_y) \hat{j} \\
 &= (-1039.5 + 1035.3) \hat{i} + (3863.7 + 4890.7) \hat{j} \\
 &= \underbrace{(-4.2)}_{R_x} \hat{i} + \underbrace{(8754.3)}_{R_y} \hat{j}
 \end{aligned}$$

Magnitude

$$|\vec{R}| = \sqrt{(R_x)^2 + (R_y)^2} = \sqrt{(-4.2)^2 + (8754.3)^2}$$

$|\vec{R}| = 8754 \text{ unit}$

Direction

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right) = \tan^{-1} \left(\frac{8754.3}{-4.2} \right)$$

$\theta = -89.97^\circ$

$\theta_+ = 180^\circ - 89.97 = 90.03^\circ$

 Almost along the dotted line ①

Summary: Ch 1 & Ch 2

- DON'T FORGET YOUR UNITS!!!!
- Make sure your numbers are in base units
- Conversion of number from one unit into another
- Significant figures
- Rule for adding/subtracting and multiplying/dividing significant figures
- Scientific notation
- Rounding a number
- Vectors have direction and magnitude
- When adding vectors, you must break them into components, i.e. x and y components

Homework 1:
**Complete Chapter 2 Learning Modules in
Webcourses.**